



DEPARTMENT OF ELECTRICAL
AND COMPUTER ENGINEERING

MARCOS VINICIUS BONFIM ROMÃO

BSc in Electrical and Computer Engineering

5G NETWORKS: DEVELOPMENT OF A 5G CORE AND RADIO ACCESS NETWORK

PRE-DISSERTATION INFORMATION

MASTER IN ELECTRICAL AND COMPUTER ENGINEERING

NOVA University Lisbon

Draft: April 3, 2026

5G NETWORKS: DEVELOPMENT OF A 5G CORE AND RADIO ACCESS NETWORK

PRE-DISSERTATION INFORMATION

MARCOS VINICIUS BONFIM ROMÃO

BSc in Electrical and Computer Engineering

Supervisor: Rodolfo Oliveira
Full Professor, NOVA University Lisbon

MASTER IN ELECTRICAL AND COMPUTER ENGINEERING

NOVA University Lisbon

Draft: April 3, 2026

CONTENTS

Acronyms	v
1 Introduction	1
1.1 Motivation	1
1.1.1 Objectives and Contributions	1
1.2 Report Structure	2
2 Elements of a 5G Network	3
2.1 Introduction to the 5G Architecture	3
2.1.1 Service-Based Architecture	4
2.1.2 Control and User Plane Separation	4
2.1.3 Network Slicing	4
2.1.4 Virtualization	5
2.1.5 Cloud-Native Functions	5
2.2 Standalone vs Non-Standalone 5G Deployments	5
2.2.1 Non-Standalone (NSA) 5G Deployments	6
2.2.2 Standalone (SA) 5G Deployments	6
2.3 5G RAN	7
2.3.1 Next Generation Radio Access Network (NG-RAN) Architecture	7
2.3.2 RAN Interfaces	9
2.3.3 RAN Features	9
2.4 5G Core	11
2.4.1 Core Network Functions	11
2.4.2 Core Interfaces	13
2.5 Existing Open-Source Software for 5G Networks	13
2.5.1 Open-Source 5G Core Implementations	14
2.5.2 Open-Source 5G RAN Implementations	15
2.5.3 UERANSIM	16
2.6 Related Works	16

3	Workplan	20
3.1	Tasks	20
3.2	Timeline	21
4	Conclusions	22
	Bibliography	23

LIST OF FIGURES

2.1	5G Network architecture diagram from the Open5GS project [19]. Used under the AGPL-3.0 license.	3
2.2	Control and User Plane Separation.	4
2.3	5G NSA vs SA Deployments	6
2.4	Next Generation Node B (gNodeB) Architecture [22]	7
2.5	5G RAN Protocol Stack [22]	8
2.6	5G RAN [22]	9
2.7	Massive MIMO antennas	10
2.8	5G Frequency Bands	10
2.9	5G Core Network Architecture [22]	11

LIST OF TABLES

2.1	5G Core Network Interfaces [20, 22]	13
2.2	Overview of different open-source 5G Core implementations.	15
2.3	Comparison of the open-source 5G RAN projects, as described by Mamushiane et al. [18].	16
2.4	Overview of the network implementations of related works.	19
3.1	Workplan Timeline	21

ACRONYMS

3GPP	3rd Generation Partnership Project (<i>pp. 6, 10, 14, 15</i>)
AF	Application Function (<i>p. 13</i>)
AMF	Access and Mobility Management Function (<i>pp. 5, 11, 13, 14, 20</i>)
API	Application Programming Interface (<i>pp. 11–13</i>)
AUSF	Authentication Server Function (<i>p. 12</i>)
AWS	Amazon Web Services (<i>p. 5</i>)
Azure	Microsoft Azure (<i>p. 5</i>)
CD/CI	Continuous Development/Continuous Integration (<i>p. 18</i>)
COTS	Commercial Off-The-Shelf (<i>pp. 10, 17–19</i>)
CU	Centralized Unit (<i>pp. 8, 9, 15, 16</i>)
CUPS	Control and User Plane Separation (<i>pp. i, 4, 11</i>)
DN	Data Network (<i>p. 13</i>)
DU	Distributed Unit (<i>pp. 8, 9, 15, 16</i>)
eMBB	enhanced Mobile Broadband (<i>p. 7</i>)
eNodeB	Evolved Node B (<i>p. 8</i>)
EPC	Evolved Packet Core (<i>pp. 3, 6, 11, 12, 14</i>)
F1-U	F1 User Plane Interface (<i>p. 9</i>)
F1AP	F1 Application Protocol (<i>p. 9</i>)
GCP	Google Cloud Platform (<i>p. 5</i>)
gNodeB	Next Generation Node B (<i>pp. iii, 7–10, 15–18</i>)
GPSDO	GPS Disciplined Oscillator (<i>p. 18</i>)
GSM	Global System for Mobile Communications (<i>p. 17</i>)
GTP-C	GPRS Tunneling Protocol - Control Plane (<i>p. 13</i>)

GTP-U	GPRS Tunneling Protocol - User Plane (p. 13)
HSS	Home Subscriber Server (p. 12)
IoT	Internet of Things (pp. 1, 4)
KPI	Key Performance Indicator (pp. 2, 20, 22)
LMF	Location Management Function (p. 13)
LPWA	Low-Power Wide-Area (p. 4)
LTE	Long-Term Evolution (pp. 10, 15, 17)
MAC	Medium Access Control (p. 8)
MIMO	Multiple Input Multiple Output (p. 10)
MME	Mobility Management Entity (p. 11)
mMTC	massive Machine Type Communications (p. 7)
mmWave	Millimeter Wave (p. 10)
MSc	Master of Science (p. 21)
N3IWF	Non-3GPP Interworking Function (p. 13)
NEF	Network Exposure Function (p. 12)
NG-C	NG Control Plane Interface (p. 13)
NG-RAN	Next Generation Radio Access Network (pp. i, 7, 8)
NGAP	NG Application Protocol (p. 13)
NR	New Radio (pp. 9, 10)
NRF	Network Repository Function (pp. 4, 12)
NSA	Non-Standalone (pp. i, 5, 6, 14, 16, 17)
NSSF	Network Slice Selection Function (p. 13)
NWDAF	Network Data Analytics Function (p. 13)
OAI	OpenAirInterface (pp. 15, 19)
OSS	Open-Source Software (pp. 13, 14, 16, 18)
OTA	Over-The-Air (p. 18)
PCF	Policy Control Function (pp. 12, 14, 20)
PCRF	Policy and Charging Rules Function (p. 12)
PDCP	Packet Data Convergence Protocol (p. 8)
PDU	Protocol Data Unit (pp. 11, 18)
PHY	Physical Layer (p. 8)
QoS	Quality of Service (pp. 8, 10, 13)

RAN	Radio Access Network (<i>pp. 1, 2, 5–7, 9, 13–18, 20, 22</i>)
RLC	Radio Link Control (<i>p. 8</i>)
RRC	RAN Control Protocol (<i>p. 8</i>)
RU	Radio Unit (<i>p. 8</i>)
SA	Standalone (<i>pp. i, 5, 6, 14, 16, 17, 22</i>)
SBA	Service-Based Architecture (<i>pp. i, 4</i>)
SCEF	Service Capability Exposure Function (<i>p. 12</i>)
SDR	Software-Defined Radio (<i>pp. 16–18, 20, 22</i>)
SMF	Session Management Function (<i>pp. 11, 13, 14, 20</i>)
SMSF	Short Message Service Function (<i>p. 13</i>)
UDM	Unified Data Management (<i>p. 12</i>)
UDR	Unified Data Repository (<i>p. 12</i>)
UE	User Equipment (<i>pp. 5, 7–13, 16–18</i>)
UPF	User Plane Function (<i>pp. 11, 13, 14, 20</i>)
URLLC	Ultra-Reliable Low-Latency Communications (<i>pp. 4, 7</i>)
VM	Virtual Machine (<i>p. 5</i>)

INTRODUCTION

1.1 Motivation

Nowadays, 5G networks are no longer just an evolution of mobile broadband communications, they are the technological foundation for a new generation of applications that demand infallible reliability, ultra-low latency and massive connectivity. From industrial automation [7, 9, 16], extended and augmented reality [17, 21, 24] to Internet of Things (IoT) applications [10, 17, 24, 26], these modern use cases require a robust network architecture that can be reconfigured dynamically, scaled up or down and overall, be tailored to meet their specific needs.

This shift has placed enormous emphasis on programmability, softwarization and virtualization across both RAN and Core components. As a result, network operators, research institutions, academic investigators and industry stakeholders are steering away from proprietary systems and looking towards open-source, cloud-native 5G solutions that can be easily adapted to different scenarios and requirements. These platforms make it possible to explore new algorithms, validate architectural choices and test emerging technologies without the hurdles imposed by closed, vendor-specific systems.

However, deploying a fully functional 5G network based solely on open-source components is a complex task, requiring a deep understanding of the 5G architecture, protocols and best practices, as well as familiarity with the plethora of available open-source frameworks, along with their constraints. This report aims to explore the available open-source solutions for 5G networks, comparing their features and drawbacks.

1.1.1 Objectives and Contributions

This subsection outlines the primary objectives that were defined for this thesis project, as well as a few expected contributions to the field of 5G networking.

Objectives:

- Elaboration of a comprehensive literature review on 5G network architectures, protocols, and open-source implementations, identifying key design challenges and opportunities in the field.

- Development of a 5G network testbed, including both RAN and Core components, using open-source software and accessible hardware.
- Evaluation of Key Performance Indicators (KPIs) of the deployed 5G network under various scenarios, such as throughput, latency, reliability and handover performance.

Expected Contributions:

- A detailed analysis of existing open-source 5G solutions, highlighting their strengths, weaknesses, and suitability for different deployment scenarios.
- A functional 5G network testbed that can be used for further research and experimentation in the field of 5G networking.

1.2 Report Structure

The structure of this document is as follows:

Chapter 2 This chapter details extensively the fundamental elements and architecture of 5G networks, going into detail about both the Radio Access Network (RAN) and Core components, along with their features, as well as current open-source solutions for implementing the architecture. An overview of the existing testbeds and experimental projects in the field of 5G networks is also provided.

Chapter 3 This chapter describes the work plan and the milestones to reach along the thesis project.

Chapter 4 This chapter concludes the document, summarizing the key points and outlining the next steps for the thesis project.

ELEMENTS OF A 5G NETWORK

2.1 Introduction to the 5G Architecture

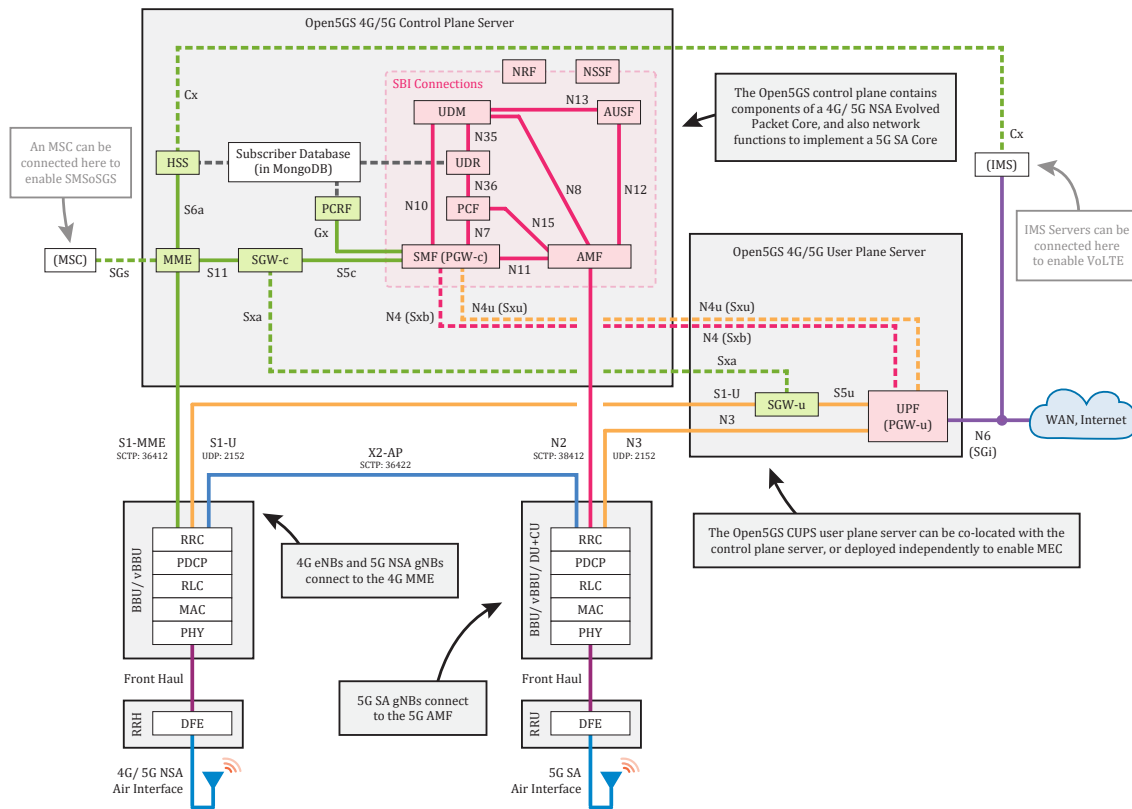


Figure 2.1: 5G Network architecture diagram from the Open5GS project [19]. Used under the AGPL-3.0 license.

The 5G architecture represents a significant evolution from previous generations of mobile networks, particularly the 4G Evolved Packet Core (EPC). Compared to the EPC, the 5G architecture is designed to be more flexible, scalable, and efficient, enabling a wide range of novel use cases and services. Key features of the 5G architecture include:

2.1.1 Service-Based Architecture

Unlike the monolithic architecture of 4G and previous generations, 5G adopts a service-based approach where network functions are designed to be modular services that can be independently configured and implemented. Network functions expose their capabilities and communicate via HTTP/2, using a standardized REST API, allowing for interoperability between services.

In this paradigm, the functions take the roles of either Service Providers or Service Consumers, depending on whether they offer or utilize specific services. The Network Repository Function (NRF), which will be developed upon further, plays a crucial role in this architecture by maintaining a registry of available network functions and their services, enabling dynamic discovery and interaction among them.

2.1.2 Control and User Plane Separation

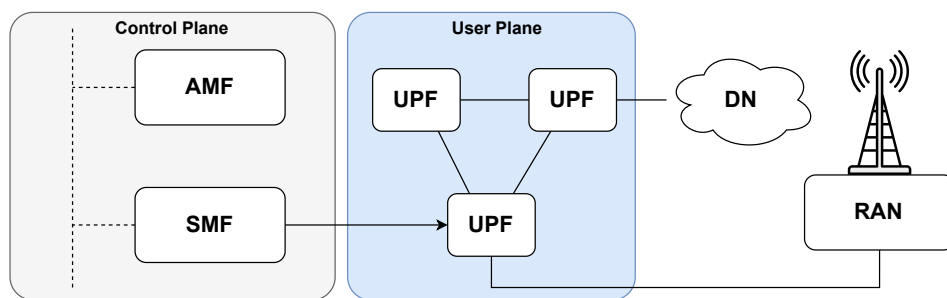


Figure 2.2: Control and User Plane Separation.

In 5G systems, the separation of the control plane and user plane is one of the fundamental design principles that sets them apart from previous generations, defining them as independent functional entities that can be deployed, scaled and managed separately, as shown on figure 2.2.

This means that network operators can dimension their networks to adapt to real network traffic patterns, deploying control plane functions in centralized locations for efficient signaling, while user plane functions can be distributed closer to the network edge to minimize service latency and hiccups, improving throughput and overall user experience.

2.1.3 Network Slicing

The concept of network slicing isn't new, but 5G develops upon it, driven by the rising popularity of odd use cases, such as Low-Power Wide-Area (LPWA) networks, IoT devices and Ultra-Reliable Low-Latency Communications (URLLC) applications.

Network slicing allows operators to create and deploy multiple virtual networks on top of a shared physical infrastructure, each optimized for specific use cases and services.

5G network slicing also allows operators to dynamically allocate resources based on the subscription and applications running on the User Equipment (UE).

In contrast to 4G EPS slicing, a 5GS network slice consists of both RAN and Core components and allows for multiple concurrent slices to be active on a single UE, with the only drawback being that the RAN and Access and Mobility Management Function (AMF) must be shared across those network slices.

2.1.4 Virtualization

Traditionally, components of a mobile network were tied to hardware appliances that only ran a specific function. This meant that software had to be pre-installed and configured for each network function, leading to inefficiencies in resource utilization and scalability.

In 5G, network functions can be virtualized, running in Virtual Machines (VMs) or containers, allowing them to share physical hardware with other functions. This approach provides greater flexibility and resource optimization, since functions are no longer tied to the hardware and can be dynamically allocated and decommissioned given current network demands.

2.1.5 Cloud-Native Functions

Building upon virtualization, 5G network functions are designed to be cloud-native, meaning they are developed from scratch and optimized to be run in cloud environments, such as public clouds (e.g. Amazon Web Services, Google Cloud Platform, Microsoft Azure, etc.) or private operator clouds.

This means that network operators can run part of their 5G network in cloud environments, taking advantage of their resilience and scalability, as well as the ability to deploy network functions closer to end-users, given their global reach, reducing network latency.

Being cloud-native, network operators can also take advantage of orchestration and automation tools, such as Kubernetes and OpenStack [18], to manage the lifecycle of network functions, deployment, scaling and healing, further enhancing the efficiency of their 5G networks.

2.2 Standalone vs Non-Standalone 5G Deployments

The adhesion to 5G technology has been a gradual process for many network operators, as the advent of 5G networks, while innovative, also meant a significant revamp in existing infrastructure, which brought along considerable costs, logistical challenges and unavoidable service disruptions. Before evolving into fully independent 5G networks, many operators opted for integrating 5G capabilities into their existing 4G infrastructure. This approach is aptly named Non-Standalone, while complete 5G networks are known as Standalone deployments. In this section, we explore the differences between these two deployment modes.

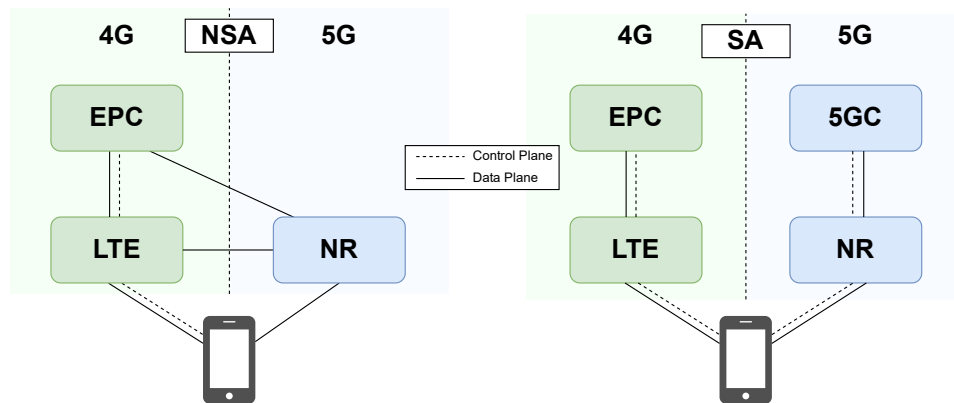


Figure 2.3: 5G NSA vs SA Deployments

2.2.1 Non-Standalone (NSA) 5G Deployments

Non-Standalone deployments were introduced as a way to accelerate the rollout of 5G services while keeping costs minimal, by leveraging existing 4G infrastructure while implementing 5G RAN components. In this architecture, the 4G EPC continues to handle control plane functions, such as mobility management and session control, but user plane data is transmitted over the 5G RAN, allowing users to benefit from the enhanced data throughput of 5G technology.

But this approach has the drawback of still relying on the 4G architecture, which means that it doesn't take advantage of the full capabilities of 5G, described before in Section 2.1. Additionally, while keeping costs low, Non-Standalone (NSA) deployments increase the complexity of the network, as operators need to manage and maintain both 4G and 5G components, leading to an increased operational overhead.

In the end, NSA deployments served as a transitional phase, allowing operators to gradually introduce 5G services while preparing for the costly, time-consuming shift to Standalone (SA) deployments.

2.2.2 Standalone (SA) 5G Deployments

Standalone deployments are the true realization of the 5G system as defined by 3GPP Release 15 [3]. In these deployments, both the Core and Radio Access Networks are built entirely on 5G technology, utilizing the full suite of 5G network functions and their capabilities. This means that all control plane and user plane functions are handled by the 5G Core network, allowing operators to reap the benefits of the 5G architecture.

Although SA deployments require a significant investment up-front, they offer clear advantages over NSA deployments, by keeping the network completely native to 5G technology. Besides all advantages, this makes them more future-proof, as they can easily adapt to new 5G releases as they are introduced, proving to be more cost-effective in the long run.

2.3 5G RAN

The 5G Radio Access Network constitutes the interface between UEs and the 5G Core network, providing the radio connectivity and resource management functions essential to mobile communication. It encompasses the physical, MAC and higher layers of the protocol stack, and enables data transmission, mobility management and service differentiation. Beyond providing radio connectivity, the 5G RAN plays a central role in enabling the diverse service categories defined for 5G networks, such as enhanced Mobile Broadband (eMBB), URLLC and massive Machine Type Communications (mMTC). To meet these requirements, the 5G RAN has evolved into a highly flexible, programmable architecture, capable of operating across a wide range of frequency bands, adapting its numerology, scheduling and resource allocation strategies dynamically.

The NG-RAN, compared to its predecessors, has evolved in the sense of softwarization and disaggregation of the RAN, reflecting in modular components and virtualized functions that can be deployed in cloud-native environments. As a result, in this generation, the RAN is not seen as a fixed hardware entity, but as a dynamic platform that interacts closely with the 5G Core network and its functions to deliver optimized connectivity and services.

This chapter provides a detailed overview of the 5G RAN architecture and features, highlighting its key components and functionalities.

2.3.1 NG-RAN Architecture

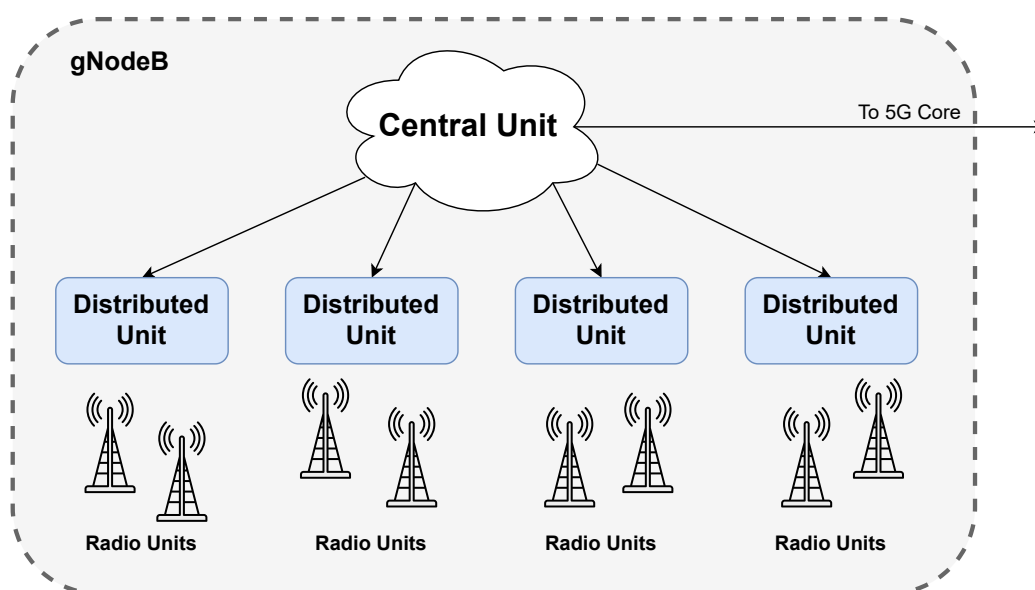


Figure 2.4: gNodeB Architecture [22]

The architecture of the NG-RAN has been fundamentally redesigned to provide higher flexibility by disaggregating the traditional base station functions into more modular components. At its core, the NG-RAN is built around the Next Generation Node B (gNodeB) (Figure 2.4), the 5G equivalent of 4G'S eNodeB, responsible for radio transmission, user-plane data forwarding and control-plane signaling with the 5G Core network.

The gNodeB is further divided into three main functional units:

Centralized Unit

This unit is responsible for higher-layer protocol processing, such as RAN Control Protocol (RRC) and Packet Data Convergence Protocol (PDCP), handling mobility, security and Quality of Service (QoS) flows, and is typically deployed in centralized locations and/or cloud environments. The Centralized Unit (CU) can manage multiple Distributed Units (DUs), coordinating resource usage across a wide area.

Distributed Unit

The DU handles lower-layer protocol processing, including Radio Link Control (RLC), Medium Access Control (MAC) and Physical Layer (PHY) functions. Multiple of these units can be connected to the same CU and they are usually deployed closer to the radio sites to minimize latency and optimize radio resource management.

Radio Unit

The Radio Units (RUs) are responsible for the radio frequency functions, including signal transmission and reception, amplification, and antenna management. These are composed of the actual antennas which interact with the UEs, along with the necessary back-end, and they are typically located at the cell site. They are connected to the DU through front-haul links.

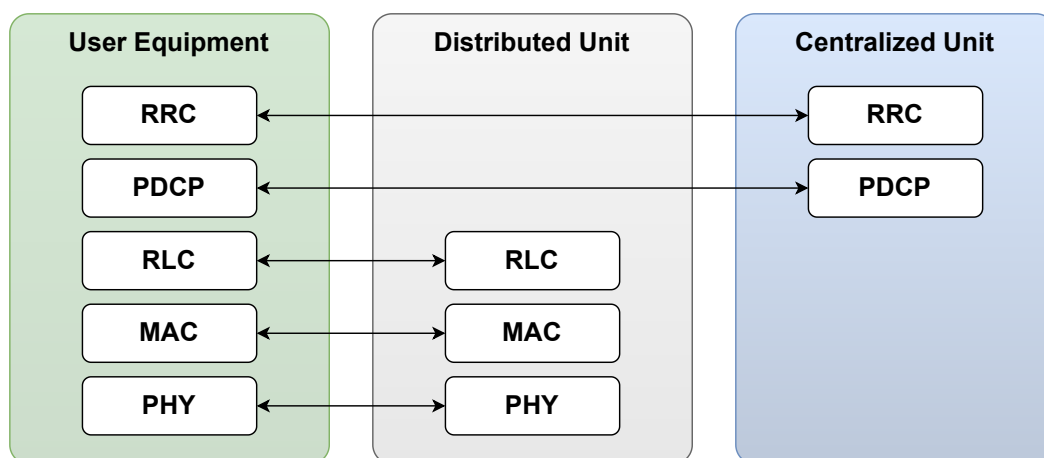


Figure 2.5: 5G RAN Protocol Stack [22]

2.3.2 RAN Interfaces

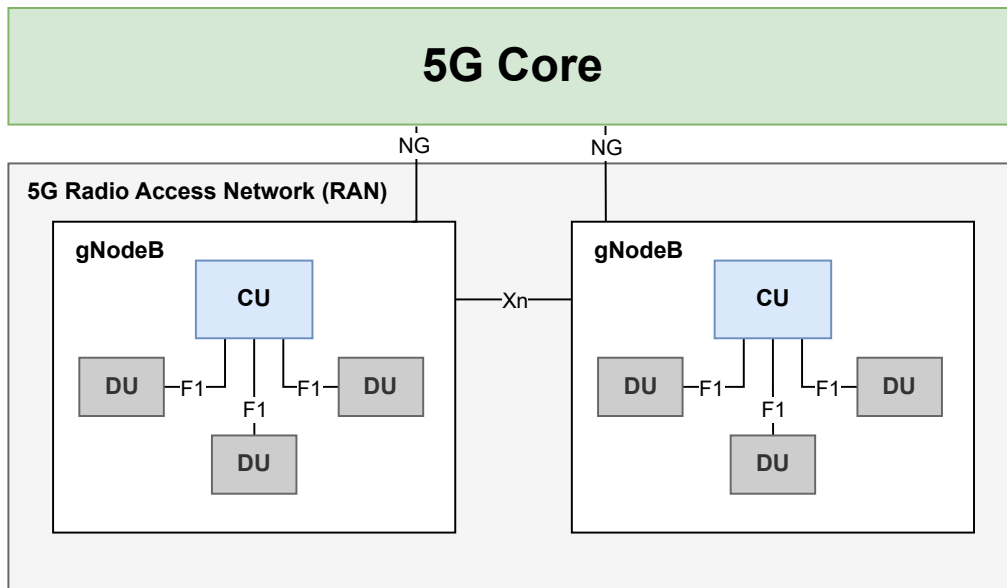


Figure 2.6: 5G RAN [22]

As depicted in Figure 2.6, the 5G RAN incorporates several key interfaces that facilitate communication between its components and with the 5G Core network. The primary interfaces include:

NG Interface

The NG interface connects the gNodeB to the 5G Core network. It is further divided into two sub-interfaces: NG-C for control plane signaling and NG-U for user plane data transfer, facilitating both control plane and user plane communications.

Xn Interface

The Xn interface connects a gNodeBs between each other, enabling inter-cell coordination and handover management. This cell also supports dual connectivity scenarios, where a UE can be simultaneously connected to multiple gNodeBs for improved performance and reliability.

F1 Interface

The F1 interface connects the DUs to the CU within the gNodeB. It is also divided into two sub-interfaces: F1 User Plane Interface (F1-U) for user plane data and F1 Application Protocol (F1AP) for control plane signaling.

2.3.3 RAN Features

5G NR develops upon pre-existing technologies, introduced in previous generations, while introducing new features to meet the ever-demanding requirements of modern mobile

communications. One feature that differs NR from its predecessors is its Multiple Input Multiple Output (MIMO) technology, introducing Massive MIMO.

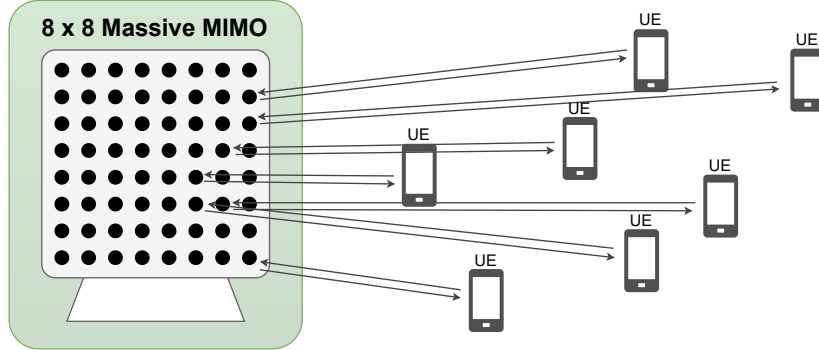


Figure 2.7: Massive MIMO antennas

In 4G LTE, MIMO capabilities were already employed, dividing the available antennas between different spatial cells to increase spectral efficiency and throughput. However, 5G New Radio (NR) takes this concept further by significantly increasing the number of antennas at the base station, often with upwards of 100+ antennas per gNodeB (Figure 2.7), allowing for User Equipments to be served simultaneously by a single node using the same time-frequency resources.

Each antenna is capable of beamforming, which focuses the radio signal directly towards the UE, rather than broadcasting it omnidirectionally. By dynamically adjusting the phase and amplitude of the signal at each antenna, the system can create highly directional beams that target specific users or cells. This not only improves signal strength and throughput, but also reduces wasted energy and interference with other devices, enhancing the QoS.

Driven by the increasing demand for higher bandwidths, 3GPP Release 17 [4] extended the operation spectrum up to 71 GHz, subdividing the frequency ranges into two categories, as shown in Figure 2.8.

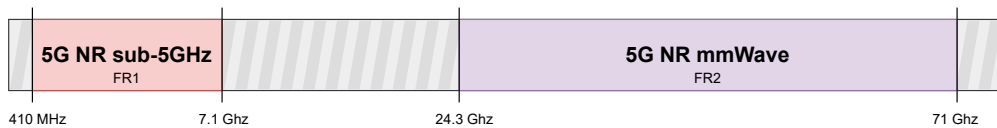


Figure 2.8: 5G Frequency Bands

- **Frequency Range 1 (FR1):** Known as the Sub-6GHz range, it encompasses frequencies from 410 MHz to 7.125 GHz.
- **Frequency Range 2 (FR2):** Known as mmWave, it covers frequencies from 24.25 GHz to 71 GHz. This range greatly increases available bandwidth and, therefore, throughput, but suffers from significant path loss. It is important to note that many COTS devices are not compatible with the frequency range.

2.4 5G Core

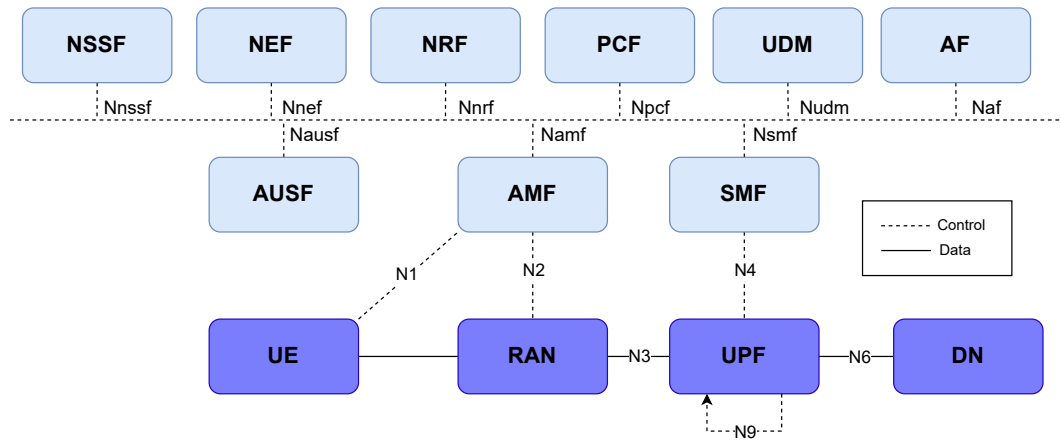


Figure 2.9: 5G Core Network Architecture [22]

As mentioned before [2.1.1], a key difference between the 5G Core network and the EPC used in 4G networks is the adoption of a service-based architecture. Network Functions are main components that form the 5G Core, and are designed to be modular and communicate with each other using standardized Application Programming Interfaces (APIs), allowing for greater flexibility, scalability, and the ability to introduce new services easily.

2.4.1 Core Network Functions

As discussed in Section 2.1.2, Control and User Plane Separation (CUPS) defines the separation of control plane and user plane functions. As defined by Rommer et al. [20], the main control plane functions are described below:

2.4.1.1 Control Plane

Access and Mobility Management Function

The AMF is a key control plane function in the 5G Core network, counterpart of the EPC's Mobility Management Entity (MME). It is responsible for handling connection and mobility management tasks for the UE. It manages registration, authentication, reachability, and mobility procedures, ensuring seamless connectivity as users move across different network areas. The AMF also interfaces with other network functions to facilitate session management and policy enforcement.

Session Management Function

The Session Management Function (SMF) is the function responsible for establishing, modifying and releasing Protocol Data Unit (PDU) sessions, determining the protocol used for communication (IPv4, IPv6, Ethernet, etc), and selecting the User Plane Function (UPF) that will handle the user plane traffic for each session. It also manages

IP address allocation for the UE and, along with the Policy Control Function (PCF), enforces policies related to session management.

Policy Control Function

Equivalent to EPC's Policy and Charging Rules Function (PCRF), the PCF is responsible for policy control and decision-making within the 5G Core network. Besides this, the PCF also has the additional responsibility of providing policy information to the UE, such as Access Network Discovery, Selection Policies and UE Route Selection Policies.

Unified Data Management & Unified Data Repository

The Unified Data Management (UDM), along with the Unified Data Repository (UDR), serve as the 5GC equivalent to EPC's Home Subscriber Server (HSS). These functions are responsible for storing and managing subscriber data and profiles. The UDR stores user subscription information, including authentication credentials, service entitlements, and policy data. The UDM serves as a front-end to the registry, interacting with other network functions, specifically with the PCF and NEF to facilitate user authentication, authorization, and mobility management.

Network Repository Function

The Network Repository Function is a central component of the 5G Core network that serves as a service registry and discovery function. It maintains a database of available network functions and which services they offer, allowing other network functions to dynamically discover and interact with each other, without the need for them to be configured beforehand. The NRF plays a crucial role in enabling the service-based architecture of the 5G Core, facilitating communication and coordination among various network functions.

Authentication Server Function

The Authentication Server Function (AUSF) is responsible for handling authentication procedures within the 5G Core network. It works in conjunction with the UDR/UDM to authenticate UEs attempting to access the network. The AUSF is also responsible for providing security parameters in roaming contexts and ensuring that authentication processes comply with 5G security standards.

Network Exposure Function

Equivalent to EPC's Service Capability Exposure Function (SCEF), the Network Exposure Function (NEF) acts as an intermediary between the 5G Core network and external application functions. It exposes network services and events, such as the location of an UE, the reachability of the UE, its roaming status and loss of connectivity can be exposed by the NEF to authorized applications and functions, enabling them to interact with the network in a secure and controlled manner. The NEF also handles API management, security, and policy enforcement for these interactions.

Other more specific control plane functions, such as the Network Slice Selection Function (NSSF), Network Data Analytics Function (NWDAF), Non-3GPP Interworking Function (N3IWF), Application Function (AF), Short Message Service Function (SMSF) and Location Management Function (LMF) are defined.

2.4.1.2 User Plane

The user plane in the 5G Core network is composed of a single main function, the User Plane Function. It is responsible for handling all the user data traffic between the RAN and external data networks. It performs packet routing and forwarding, QoS enforcement, traffic shaping, as well as usage reporting and, in some cases, lawful interception. Multiple UPFs can be deployed in series, co-existing within the same 5G Core network, enabling distributed architectures and facilitating scalability if necessary.

2.4.2 Core Interfaces

Information exchange inside the 5G Core happens through a common access medium using APIs but, for communication with outside components, it relies on a set of interfaces that define how UEs, the RAN and the various network functions exchange information. These interfaces are designed to support the service-based architecture of the 5G Core, communicating directly with network functions. The main interfaces, as seen on figure 2.9, are summarized in Table 2.1.

Interface	Connections	Protocol
N1	UE ↔ AMF	NAS over NG-C
N2	RAN ↔ AMF	NGAP
N3	RAN ↔ UPF	GTP-U
N4	SMF ↔ UPF	GTP-C
N6	UPF ↔ external DNs	IP-based Protocols
N9	UPFs ↔ UPFs	GTP-U

Table 2.1: 5G Core Network Interfaces [20, 22]

2.5 Existing Open-Source Software for 5G Networks

Overall, Open-Source Software (OSS) has been a growing tendency in the computer engineering field, and the telecommunications sector is no exception. The adoption of OSS for 5G networks has been driven by the need for cost-effective, flexible and customizable solutions that can adapt to the rapidly evolving landscape of mobile communications.

These tools play a crucial role in research, prototyping and, in some cases, even production deployments of 5G networks, as they allow developers to freely experiment with 5G architectures without relying on expensive proprietary solutions. The existing

suite of open-source projects already spans both the 5G Core and RAN, as well as other tools for support testing, evaluating performance and validation.

2.5.1 Open-Source 5G Core Implementations

On the core side, many projects have emerged to provide an accessible way to deploy a complete 5G core network using OSS components, with varying levels of maturity. Some of the most notable projects will be described below, with a brief overview on Table 2.2:

2.5.1.1 Open5GS

Open5GS [19] is an open-source implementation of a 5G Core network, written in C language. Out-of-the-box, it provides a set of both 5G Core network functions, such as control plane functions (AMF, SMF, PCF, etc.) and the user plane function (UPF), as well as 4G EPC functions for NSA implementations and backwards compatibility. It focuses on being lightweight and easy to deploy, containing extensive documentation which guides a developer from installation to configuration and, finally, operation. Each network function in Open5GS is defined in a separate configuration file, making it straightforward to modify individual components without affecting the entire system. Open5GS is one of the most widely adopted open-source 5G Core implementations, as seen in [8, 15, 18, 23], making it a popular choice for research and prototyping of 5G networks.

2.5.1.2 Free5GC

Free5GC [12] is another open-source project that offers an implementation of the 5G Core, written in Go, which integrates well with web-services. It provides a full suite of 5G Core network functions, as according to 3GPP Release 15 [3]. Although not as popular as other deployments, Free5GC has been used in academia, as seen in [15], and has an active community that contributes to its development and maintenance.

2.5.1.3 Magma

Initially developed by Facebook, Magma [11] is a project that provides 2G, 3G, 4G and 5G Core network implementations, made open-source in 2019. Despite being relatively new, it has seen a spike in contributors and users, mainly due to being designed from scratch to be implemented in poor, rural areas, where connectivity is scarce. Within their project, they offer a full 5G SA Core network implementation, written mainly in C++, Go and Python.

While not as widely adopted as other solutions, due to its novelty and lack of extensive documentation, Magma has been used in research, as seen in [14].

2.5.1.4 OpenAirInterface Core Network

The OpenAirInterface Software Alliance, composed by major global players (such as Nokia, Vodafone or Ericsson), provides a 5G Core [5] implementation, mainly maintained by its community. It is written in C++ and provides a complete set of 5G Core network functions, based on containers, making it easy to deploy and manage. The project is well-documented, has many tutorials and examples and has an active community. It is widely used in academia and research, as seen in [8, 15, 25]

Project	Programming Language	Supported Generations	Maturity level
Open5GS	C	4G/5G	Most widely adopted open-source 5G Core implementation, with extensive documentation and community support. Used in multiple research projects.
Free5GC	Go	5G	Well-maintained project with a focus on web-services integration. Less popular than Open5GS but still used in academia.
Magma	C++, Python	Go, 2G/3G/4G/5G	Newer project with a focus on rural connectivity. Growing community and adoption, but less mature than other solutions.
OAI Core	C++	4G/5G	Backed by major industry players, well-documented and widely used in academia. Container-based deployment for ease of management.

Table 2.2: Overview of different open-source 5G Core implementations.

2.5.2 Open-Source 5G RAN Implementations

On the RAN side, many open-source projects have also emerged to provide 5G RAN implementations, but two key players stand out from the rest, srsRAN and OpenAirInterface RAN, due to their maturity, extensive documentation and active communities. These projects are described below, followed by a brief comparison on Table 2.3.

2.5.2.1 srsRAN

srsRAN [1] is an open-source software suite that provides a complete implementation of the 5G RAN, as defined by 3rd Generation Partnership Project (3GPP). It is written in both C and C++ languages and is designed to be modular, allowing the developer to customize both the CU and DU components of the gNodeB separately.

It supports all FR1 bands, FDD/TDD modes, all bandwidths up to 100 MHz TDD and 50 MHz FDD, 15/30 kHz subcarrier spacing, 256-QAM modulation, Network Slicing, among other features. srsRAN integrates well with Open5GS and is widely adopted in both academia and industry, as seen in [15, 18, 23, 25].

The srsRAN project also provides 4G LTE support through its srsRAN 4G project [2], allowing for backwards compatibility with existing 4G networks.

2.5.2.2 OpenAirInterface

Along with a Core Network, the OpenAirInterface Software Alliance also provides a 5G RAN implementation [6], implementing full gNodeB and UE functionalities. Also written in C and C++, it is the most complete 5G open-source RAN stack, supporting both NSA and SA deployments, monolithic gNodeB as well as disaggregated CU/DU architectures and FR2 frequency band support. Just like srsRAN, it is widely used in academia, as seen in [8, 15].

Feature	srsRAN	OpenAirInterface RAN
Frequency Bands Supported	All FR1 (sub-6 GHz) bands	All FR1 and FR2 bands
Deployments supported	NSA and SA	NSA and SA
Duplex Mode	FDD/TDD	FDD/TDD
Sub-Carrier Spacing	15/30 kHz	15 and 30 kHz (FR1), 120 kHz (FR2)
Bandwidth (BW)	10, 20, 40, 80, 100 MHz	10, 20, 40, 80, 100 MHz
Programming Language	C & C++	C & C++
Antenna Configuration	SISO	SISO and MIMO
Documentation & Support	Extensive	Fair
Deployment Complexity	Easier and faster to deploy	Complex to deploy

Table 2.3: Comparison of the open-source 5G RAN projects, as described by Mamushiane et al. [18].

2.5.3 UERANSIM

UERANSIM [13] is not a real RAN implementation, but rather a 5G UE and RAN simulator, written in C++. It can be used to test 5G Core networks without the need for Software-Defined Radios (SDRs). It supports both NSA and SA deployments, and can simulate multiple UEs simultaneously.

2.6 Related Works

The deployment of 5G networks using OSS has already become an area of research, with many researchers having explored the feasibility of deploying 5G networks using open-source components. Very different approaches have been taken, from small-scale testbeds to complex environments integrated with commercial equipment. In this section, a discuss of some relevant works in this area will be presented, followed by an overview of said word on table 2.4. Most of these works were sourced from srsRAN’s Research tab, made available on their website.

Tío et al. [23] focused on implementing a complete 5G SA network testbed and evaluated its performance using different UEs. Their approach consisted of using Open5GS for the

5G Core network and srsRAN for the RAN, deployed on USRP X310 and B200 SDRs. To evaluate the performance of their testbed, they used srsUE, a 5G UE implementation developed by the srsRAN project, Commercial Off-The-Shelf UEs and a Sixfab 5G HAT mounted on a Raspberry Pi.

Their results showed that the testbed was able to provide connectivity to the UEs and achieve average data rates of up to 68.6 Mbps downlink. Their testbed was also able to connect to the internet and, using Speedtest by Ookla, achieve downlink speeds upwards of 20.1 Mbps. They also performed a series of qualitative tests: package downloads through apt, video streaming on Youtube and video calls between UEs, with varying degrees of success.

On a similar vein to the previous work, Vázquez et al. [25] too developed a complete 5G SA testbed, named "MAQ5G". Their approach relies solely on the OpenAirInterface project, deploying both their 5G Core and RAN software on USRP X310 and B210 SDRs, with some modifications to the gNodeB scheduler's code.

Although similar, the evaluation done on this work shines a bigger light on the radio performance, measuring the throughput through different means of transmission and RAN configurations, using two COTS devices outfitted with writable SIM cards. The authors also evaluated the RAN host system's performance under different traffic loads, measuring CPU usage.

Results showed that their testbed was able to provide connectivity to the UEs and was able to achieve a maximum downlink throughput of 149 Mbps, although they also observed that, under high traffic loads, the host system's behaviour became unreliable, leading to loss of connection for the UEs, being the solution a full UE reboot.

Mamushiane et al. [18] also developed a 5G SA testbed using Open5GS and srsRAN. Their report shines a bigger light on the network development, detailing extensively their configuration decisions and the necessary steps to deploy the network successfully. They also provide a guide for troubleshooting network issues.

While not being the central topic of their work, they evaluated the performance of their testbed, presenting a gNodeB trace showing perfect conditions for the connection, with 0 to 1 packets dropped on downlink and no packets dropped on uplink.

Shifting their focus towards containerization, orchestration and the complexity of managing different software stacks, Horstmann et al. [15] developed a complete general-purpose network framework named Open5GCube, able to deploy both 5G SA and NSA deployments modularly, as well as 4G LTE and 2G GSM networks.

Their approach focuses on containerization using Docker Containers, allowing the developers to interchange between different implementations of both the 5G Core and RAN components seamlessly. To orchestrate the deployment of the network, they developed their own orchestration script.

Their testbed implements Open5GS, Free5GC and OAI Core as 5G Cores, and srsRAN and OpenAirInterface RAN as RAN implementations, built upon two USRP X310 SDRs using a 10 MHz GPSDO for synchronization.

To evaluate their framework, they used several Commercial Off-The-Shelf UEs, with different Operating Systems and network chips, and a Quectel RM500Q-GL modem. Results showed that their framework was able to provide 2G and 4G connectivity to all UEs, while 5G performance varied between the different UEs, with some being able to establish connection seamlessly to the network, while others could only connect to the RAN but not initiate a PDU session.

Integrating CD/CI and automation tools, Bonati et al. [8] presented "5G-CT", a framework focused on automated deployment and Over-The-Air (OTA) testing. Their network consisted of a Open5GS core, a 5G Networks core and OpenAirInterface RAN, deployed at a Northeastern University lab.

Their work stands out due to the tools they employed: OpenShift, a Kubernetes cluster where the Core and RAN components would be deployed, and GitOps workflows, the automation tool used to orchestrate the deployment of the network. By using these tools, they automated the provisioning of their 5G networks, effectively streamlining the setup and guaranteeing a consistent and reproducible deployment process.

To demonstrate the capabilities of their framework, they conducted a series of tests over the course of nine months. During this period, the framework executed a daily operation cycle, automatically deploying the network, connecting the UE to the network and exchanging UDP traffic between them for 6 hours at varying data rates. In the end, the framework would report the results to another OpenShift pod, where they would be analyzed, and tear down the network.

Results showed that their framework was able to setup the network successfully and tests were ran as planned, presenting little to no packet-loss.

While these works focus on OSS implementations using SDRs, Zu et al. [27] bridged the gap between OSS and commercial applications, focusing on the realism and the scale of real-world deployments. Their approach consisted on implementing a 5G core using Open5GS along with a proprietary Ericsson RAN, deployed on a 30km wireless lab in City of Ames, Iowa.

Their lab was composed of 4 gNodeBs and 30+ UE sites equipped with Quectel radios, scattered across residential areas, commercial buildings and farm lands, providing a diverse range of environments to test the network's performance. Results show that their network was able to provide connectivity to all UEs and achieve throughput speeds around 320 Mbps, thus proving that an open-source 5G Core can reliably sustain carrier-grade performance and real-world deployments.

Paper	Core	RAN	UE	Containerization	Antenna Config	SDR
[8]	OAI, Open5GS, A5G	OAI	Sierra Wireless Radio	OpenShift and GitOps workflow		8x USRP X310/X410
[15]	Open5GS, Free5GC, srsRAN, OAI	OAI, srsRAN	COTS, Modem	Docker Containers		USRP X310
[18]	Open5GS	srsRAN	COTS, Emulator	OpenStack on Docker Containers	MIMO	NI-2944R
[23]	Open5GS	srsRAN	COTS, 5G HAT, Emulator	Docker Containers	MIMO	USRP X310/B200
[25]	OAI	OAI	COTS	Docker Containers	SISO (custom scheduler)	USRP X310/B210
[27]	Open5GS	Ericsson	Quectel modem	N/A	Massive MIMO	Ericsson proprietary

Table 2.4: Overview of the network implementations of related works.

WORKPLAN

The workplan of this thesis project is structured into phases as follows.

3.1 Tasks

1. Literature Review/State-of-the-Art Survey

This task consists of studying key chapters of reference books to understand the current trends, methodologies and challenges in 5G networks, both RAN and Core, as well as reviewing existing research papers and articles on 5G network deployments and performance evaluation techniques.

2. Deployment of 5G RAN

This task involves deploying the 5G RAN using software-defined radio platforms and open-source tools, such as srsLTE and the Nuand bladeRF xA4 SDR. The RAN will be configured and tested to ensure correct operation and connectivity with the 5G Core network.

3. Deployment of 5G Core Network

Using the open-source platform Open5gs, a 5G Core will be deployed, integrating it with the previously deployed RAN and verifying end-to-end connectivity. Essential network functions, such as AMF, SMF, UPF and PCF, will be configured and validated.

4. Performance Evaluation

This phase focuses on evaluating the performance of the deployed 5G network. KPIs such as throughput, latency, reliability and handover performance will be measured using appropriate testing tools and methodologies. The results will be analyzed to assess the network's performance under various conditions, as well as to identify potential bottlenecks and validate the network setup against expected standards and theoretical performance.

5. Analysis and Reporting

In this final task, all findings, design choices, deployment procedures and performance evaluation results will be documented in the MSc thesis draft.

3.2 Timeline

The workplan is structured over a period of nine months, with each task allocated a specific timeframe as follows:

Task	Month								
	1	2	3	4	5	6	7	8	9
Literature Review /State-of-the-Art Survey									
Deployment of 5G RAN									
Deployment of 5G Core Network									
Performance Evaluation									
Analysis and Reporting									

Table 3.1: Workplan Timeline

CONCLUSIONS

This thesis aims to implement a fully functional 5G SA network, including both RAN and Core elements, built entirely with open-source components and deployed on commodity hardware. The main goals are to deploy the network in a testbed environment, logging the necessary steps of development and configurations to allow for reproducibility, and to evaluate its performance under different scenarios, measuring KPIs.

A review of the current landscape shows that many projects around the implementation of open-source 5G networks have been carried out in recent years, employing several different tools and providing extensive documentation and analysis of results. Building upon this existing body of work, this thesis will focus on deploying a 5G testbed tailored to meet the specific requirements of our environment, while also contributing to the broader research community, where possible, by sharing insights and findings.

Moving forward, the next steps of this thesis will follow the methodology outlined in Chapter 3, starting with the setup of the 5G RAN and Core components, followed by the setup and fine-tuning of the SDR hardware. Finally, the performance evaluation will be conducted, gathering Key Performance Indicators values and concluding with an analysis of the results obtained.

BIBLIOGRAPHY

- [1] S. R. S. (SRS). *srsRAN Project: Open Source O-RAN 5G CU/DU Solution*. URL: https://github.com/srsran/srsRAN_Project (cit. on p. 15).
- [2] S. R. S. (SRS). *srsRAN: Open Source 4G Software Radio Suite*. URL: https://github.com/srsran/srsran_4g (cit. on p. 15).
- [3] 3rd Generation Partnership Project (3GPP). *Release 15*. <https://www.3gpp.org/specifications-technologies/releases/release-15>. Accessed: 2026-01-27. 2018 (cit. on pp. 6, 14).
- [4] 3rd Generation Partnership Project (3GPP). *Release 17*. <https://www.3gpp.org/specifications-technologies/releases/release-17>. Accessed: 2026-01-27. 2022 (cit. on p. 10).
- [5] O. S. Alliance. *oai-cn5g-fed: Federation of the OpenAir CN 5G Repositories*. URL: <https://gitlab.eurecom.fr/oai/cn5g/oai-cn5g-fed> (cit. on p. 15).
- [6] O. S. Alliance. *OpenAirInterface5G: Open Source 4G/5G RAN and Core Network*. URL: <https://gitlab.eurecom.fr/oai/openairinterface5g> (cit. on p. 16).
- [7] J. Ansari et al. “Performance of 5G Trials for Industrial Automation”. In: *Electronics* 11.3 (2022). ISSN: 2079-9292. DOI: [10.3390/electronics11030412](https://doi.org/10.3390/electronics11030412) (cit. on p. 1).
- [8] L. Bonati et al. “5G-CT: Automated Deployment and Over-the-Air Testing of End-to-End Open Radio Access Networks”. In: *IEEE Communications Magazine* 63.1 (2025), pp. 155–162. DOI: [10.1109/MCOM.001.2300675](https://doi.org/10.1109/MCOM.001.2300675) (cit. on pp. 14–16, 18, 19).
- [9] G. Brown. *Ultra-Reliable Low-Latency 5G for Industrial Automation. A Heavy Reading White Paper Produced for Qualcomm Inc.* White Paper. Accessed: 2026-01-20. Heavy Reading, 2018. URL: <https://www.qualcomm.com> (cit. on p. 1).
- [10] W. Ejaz et al. “Internet of Things (IoT) in 5G Wireless Communications”. In: *IEEE Access* 4 (2016), pp. 10310–10314. DOI: [10.1109/ACCESS.2016.2646120](https://doi.org/10.1109/ACCESS.2016.2646120) (cit. on p. 1).
- [11] Facebook. *Magma: Open Source Software Platform for Building Carrier-Grade Networks*. URL: <https://github.com/magma/magma> (cit. on p. 14).

- [12] *free5GC: Open Source 5G Core Network Based on 3GPP Release 15*. URL: <https://github.com/free5gc/free5gc> (cit. on p. 14).
- [13] A. Güngör. *UERANSIM: Open Source 5G UE and RAN (gNodeB) Simulator*. URL: <https://github.com/aligungr/UERANSIM> (cit. on p. 16).
- [14] S. Hasan et al. “Building Flexible, Low-Cost Wireless Access Networks with Magma”. In: *GetMobile: Mobile Comp. and Comm.* 27.3 (2023-11), pp. 40–47. ISSN: 2375-0529. DOI: [10.1145/3631588.3631599](https://doi.org/10.1145/3631588.3631599) (cit. on p. 14).
- [15] T. Horstmann et al. *open5Gcube: A Modular and Usable Framework for Mobile Network Laboratories*. 2025. arXiv: [2505.14501](https://arxiv.org/abs/2505.14501) [cs.NI] (cit. on pp. 14–17, 19).
- [16] K. Khujamatov et al. “Existing Technologies and Solutions in 5G-Enabled IoT for Industrial Automation”. In: *Blockchain for 5G-Enabled IoT: The new wave for Industrial Automation*. Ed. by S. Tanwar. Cham: Springer International Publishing, 2021, pp. 181–221. ISBN: 978-3-030-67490-8. DOI: [10.1007/978-3-030-67490-8_8](https://doi.org/10.1007/978-3-030-67490-8_8) (cit. on p. 1).
- [17] Z. Lv, J. Lloret, and H. Song. “Internet of Things and augmented reality in the age of 5G”. In: *Computer Communications* 164 (2020), pp. 158–161. ISSN: 0140-3664. DOI: <https://doi.org/10.1016/j.comcom.2020.08.019> (cit. on p. 1).
- [18] L. Mamushiane et al. “Deploying a Stable 5G SA Testbed Using srsRAN and Open5GS: UE Integration and Troubleshooting Towards Network Slicing”. In: *2023 International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD)*. 2023, pp. 1–10. DOI: [10.1109/icABCD59051.2023.10220512](https://doi.org/10.1109/icABCD59051.2023.10220512) (cit. on pp. 5, 14–17, 19).
- [19] *Open5GS: Open Source 5G Core Project*. AGPL-3.0 License. URL: <https://github.com/open5gs/open5gs> (cit. on pp. 3, 14).
- [20] S. Rommer et al. *5G CORE NETWORKS*. Ed. by Elsevier. Academic Press, 2020 (cit. on pp. 11, 13).
- [21] Y. Siriwardhana et al. “A Survey on Mobile Augmented Reality With 5G Mobile Edge Computing: Architectures, Applications, and Technical Aspects”. In: *IEEE Communications Surveys & Tutorials* 23.2 (2021), pp. 1160–1192. DOI: [10.1109/COMST.2021.3061981](https://doi.org/10.1109/COMST.2021.3061981) (cit. on p. 1).
- [22] S. Sirotkin. *5G Radio Access Network Architecture: The Dark Side of 5G*. Ed. by I. Press. Wiley, 2021 (cit. on pp. 7–9, 11, 13).
- [23] B. Tío et al. *Exploring 5G with Open Source Solutions: Functional Testbed and User Equipment Evaluation*. Tech. rep. Facultad de Ingeniería, Universidad de la República, 2025. URL: <https://case.ar/wp-content/uploads/2025/07/Exploring-5G-with-Open-Source-Solutions.pdf> (cit. on pp. 14–16, 19).

- [24] M. Torres Vega et al. "Immersive Interconnected Virtual and Augmented Reality: A 5G and IoT Perspective". In: *Journal of Network and Systems Management* 28.4 (2020), pp. 796–826. ISSN: 1573-7705. DOI: [10.1007/s10922-020-09545-w](https://doi.org/10.1007/s10922-020-09545-w) (cit. on p. 1).
- [25] P. Vázquez et al. "MAQ5G: Deployment of a Complete 5G Standalone Network Testbed for Testing and Development". In: *2024 XVI Congreso de Tecnología, Aprendizaje y Enseñanza de la Electrónica (TAAE)*. 2024, pp. 1–7. DOI: [10.1109/TAAE59541.2024.10604930](https://doi.org/10.1109/TAAE59541.2024.10604930) (cit. on pp. 15, 17, 19).
- [26] D. Wang et al. "From IoT to 5G I-IoT: The Next Generation IoT-Based Intelligent Algorithms and 5G Technologies". In: *IEEE Communications Magazine* 56.10 (2018), pp. 114–120. DOI: [10.1109/MCOM.2018.1701310](https://doi.org/10.1109/MCOM.2018.1701310) (cit. on p. 1).
- [27] G. Zu et al. "Real-World Integration and Evaluation of Open-Source 5G Core with Commercial RAN". In: *2025 IEEE Military Communications Conference (MILCOM)*. 2025-10, pp. 1296–1301. DOI: [10.1109/MILCOM64451.2025.11309851](https://doi.org/10.1109/MILCOM64451.2025.11309851) (cit. on pp. 18, 19).